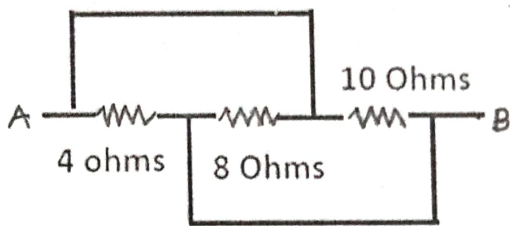
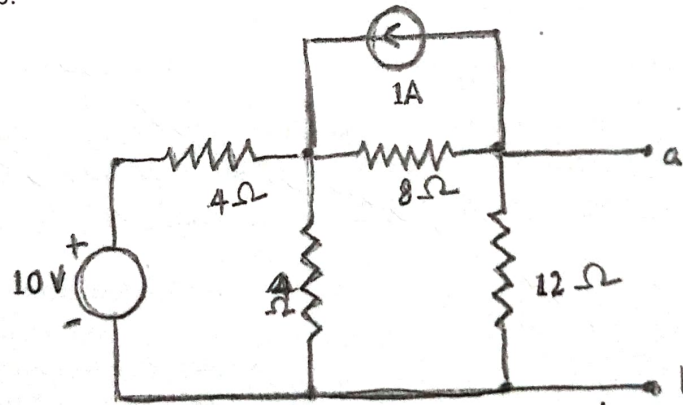
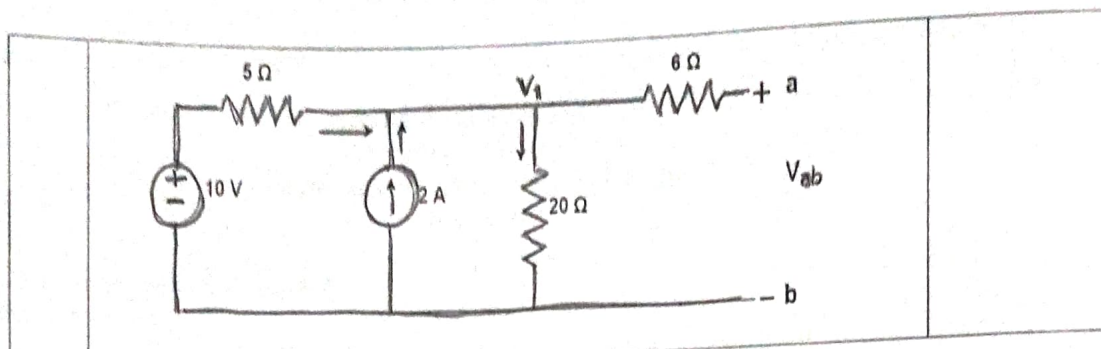


END TERM EXAMINATION
(CBCS)(SUBJECTIVE TYPE)
B. Tech (REAPPEAR), 2nd Semester
(Dec 24)

Subject Code: BEC 110	Subject: Basic Electrical Engineering
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)	CO Mapping
	(a) Differentiate between dependent and independent source.		CO1
	(b) Find the equivalent resistance between A and B.		CO2
			
	(c) Two impedance $Z=2+j3$ and $Z=5-j3$ are connected in parallel. What is the equivalent impedance of the combination?		CO2
	(d) An alternating current is given by the equation $i=10 \sin 314t$. Find the RMS value of current, frequency, time period and instantaneous value of current at 0.01 s.		CO2
	(e) How will you use a PMMC instrument which gives full-scale deflection at 50mV potential difference and 10mA current (i) ammeter 0-10 A range (ii) voltmeter 0-250 V range.		CO2
	(f) How the damping is provided in Moving Iron instruments?		CO1
	(g) Explain Fleming's right hand rule.		CO1
	(h) Derive the emf equation of transformer.		CO1
UNIT-I			CO Mapping
Q2	Find the Thevenin's Equivalent of the circuit across the terminal a and b.	(10)	CO3
			
Q3	Find out the value of load which must be connected across terminal a and b so that maximum power transfer takes place through this load and also find the value of maximum power transferred.	(10)	CO3



UNIT-II			CO Mapping
Q4	A series resonance network consisting of a resistor of 40Ω , a capacitor of $5\mu\text{F}$ and an inductor of 40mH is connected across a sinusoidal supply voltage which has a constant output of 9 volts at all frequencies. Draw the phasor diagram and find the (i) value of current at resonance (ii) The voltage across the inductor and capacitor at resonance. (iii) The quality factor and the bandwidth of the circuit.	(10)	CO3
Q5	A parallel resonance network consisting of a resistor and inductor in series with an inductor in parallel. Derive the following (i) The resonance frequency (ii) The quality factor (iii) The bandwidth of the circuit (iv) The circuit current at resonance and	(10)	CO2
UNIT-III			CO Mapping
Q6	Compare Moving Iron and Moving Coil instrument in terms of merits and demerits. How the range of moving coil instrument can be extended?	(10)	CO2
Q7	Draw the circuit diagram of two wattmeter method for measuring the power in a three-phase star connected load. Explain with the help of phasor diagram how the wattmeter reading will be measured. Find the expression for power factor for three-phase power measurement using Two-Wattmeter method.	(10)	CO4
UNIT-IV			CO Mapping
Q8	Draw the construction of DC machine and explain the working.	(10)	CO2
Q9	Draw the equivalent circuit of a transformer referred to primary side. Explain the significance of each term in it.	(10)	CO2